

sage on the LCD. During charging, the battery voltage is continuously monitored. When the voltage reaches 14.0V, the microcontroller interrupts the charging current by energising the relay, which is connected to MOSFET BS170 (T2), and starts a 5-minute timer. During this stage, the LCD shows "Battery Full."

After five minutes, the relay reconnects the panel to the battery. This way, the charging current is pulsed at the intervals of five minutes and the cycle repeats until the panel voltage is present.

When the panel voltage falls below the zener diode (ZD1) voltage of the dusk-to-dawn sensor, the microcontroller senses this and activates the load by switching on MOSFET T3 via optocoupler IC3 and "Load On" message is displayed.

In this mode, the microcontroller monitors for low battery. When the battery voltage drops below 10 volts, the microcontroller turns off the load by switching off MOSFET T3 and "Battery Low—Load Off" message is displayed.

Normally, when the load is switched off, the battery voltage tends to rise back and the load oscillates between 'on' and 'off' states. To avoid this, the

microcontroller employs a hysteresis control by entering into a 'lock' mode during low-battery state and comes out of the lock mode when the dusk-to-dawn sensor receives the panel voltage (the next morning). During lock mode, the microcontroller keeps converting the ADC value and displays the battery voltage on the LCD.

Construction and testing

Pin configurations of transistor BC547, MOSFET BS170 and MOSFET IRF540 are shown in Fig. 2. An actual-size, single-side PCB for the microcontroller-based solar charger is shown in Fig. 3 and its component layout in Fig. 4. Wire the circuit on the PCB. Prior to inserting the programmed microcontroller into the PCB, check for soldering mistakes like shorts, and for proper connections using a multimeter. Mount power MOSFET IRF540N on a suitable heat-sink. Before switching on the controller unit, connect the leads of the battery, load and solar panel at appropriate places on the board.

Switch on the unit and the message "Solar Charge Controller-EFY" is displayed on the LCD for two seconds. The system status messages are displayed on line 1 of the LCD and the battery voltage is displayed on line 2.

A small graphic representing the battery status is also displayed on line 2 of the LCD.

EFY note. 1. If the unit is switched on without the solar panel connected, the "Battery Low—Load Off" message is displayed irrespective of the battery voltage. The display changes to "Charging" as soon as the panel is connected.

2. There will be slight variation in the voltage displayed because of the tolerance levels of potential-divider resistors in the ADC section and V_{ref} of the ADC being directly connected to V_{CC} (the output of 7805 has an accuracy of 2-5 per cent) instead of dedicated temperature-compensated voltage reference.

Software

The source program for the project is written in Assembly language and assembled using Metalink's ASM51 assembler, which is freely available on the Internet for download. It is well commented for easy understanding and works as per the flow-chart shown in Fig. 5. The hex file 'solar.hex' is to be burnt into the microcontroller.

EFY note. The source code and other relevant files of this article are included in this month's EFY-CD.

SOLAR.ASM

```
SMOD51
; LCD 4-BIT MODE CONNECTIONS
RS EQU P1.7 ; LCD REGISTER SELECT
LINE
EN EQU P1.6 ; LCD ENABLE LINE
DB4 EQU P1.5 ;
DB5 EQU P1.4 ;
DB6 EQU P1.3 ;
DB7 EQU P1.2 ;
; ADC0831 CONNECTIONS
CS EQU P3.0
CLK EQU P3.1
DO EQU P3.2
; INPUT & OUTPUT
DYI EQU P3.4 ; SOLAR PANEL VOLTAGE
SENSOR
CHG_RL EQU P3.5 ; CHARGING CONTROL
RELAY
LD_RL EQU P3.7 ; LOAD CONTROL RELAY
DSEG
ORG 0020H
VAL1: DS 1
VAL2: DS 1
VAL3: DS 1
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```
ADC_VAL: DS 1
BUF: DS 1
CNT1: DS 1
CNT2: DS 1
IME: DS 1
FLAGS: DS 1
OCF BIT FLAGS.0 ; OVER CHARGE FLAG
LBF BIT FLAGS.1 ; LOW BATT FLAG
CSEG
ORG 0000H
JMP MAIN
ORG 000BH ;Timer Interrupt0
JMP COUNTDOWN
MAIN: MOV SP,#50H
MOV P3,#0FFH
MOV P1,#0FFH
CLR CHG_RL
CLR LD_RL
LCALL PWR_DELAY
LCALL INIT
SETB CLK
SETB DO
SETB CS
SETB DYI
MOV VAL1,#00H
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```
MOV VAL2,#00H
MOV VAL3,#00H
MOV FLAGS,#00H
LOADCHAR: MOV BUF,#40H
LCALL CMD
MOV DPTR,#CHAR
REP: CLR A
MOVC A,8A+DPTR
JZ SCREEN1
MOV BUF,A
LCALL DAT
INC DPTR
SJMP REP
SCREEN1: MOV BUF,#80H
LCALL CMD
MOV DPTR,#MSG1
HERE: CLR A
MOVC A,8A+DPTR
JZ NEXT
MOV BUF,A
LCALL DAT
INC DPTR
SJMP HERE
NEXT: MOV BUF,#0C0H
LCALL CMD
MOV DPTR,#MSG2
HERE1: CLR A
```